SDLC Using Agentic AI

Building an Adaptive AI Software Engineering Organization

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What is an AI Agent?

AI agents can encompass a wide range of functions beyond natural language processing including decision-making, problem-solving, interacting with external environments and performing actions.

AI agents solve complex tasks across enterprise applications, including software design, IT automation, code generation and conversational assistance.

They use the advanced natural language processing techniques of large language models (LLMs) to comprehend and respond to user inputs step-by-step and determine when to call on external tools. [17]

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Agent = LLM + Memory + Prompt Template + Tools
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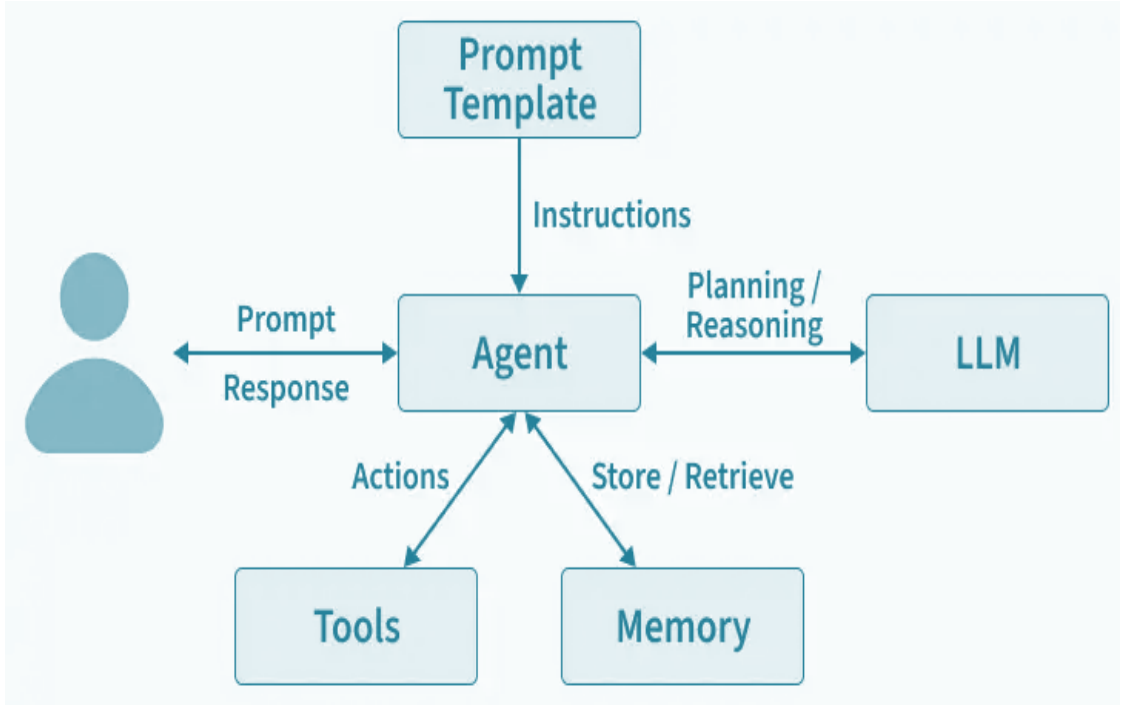
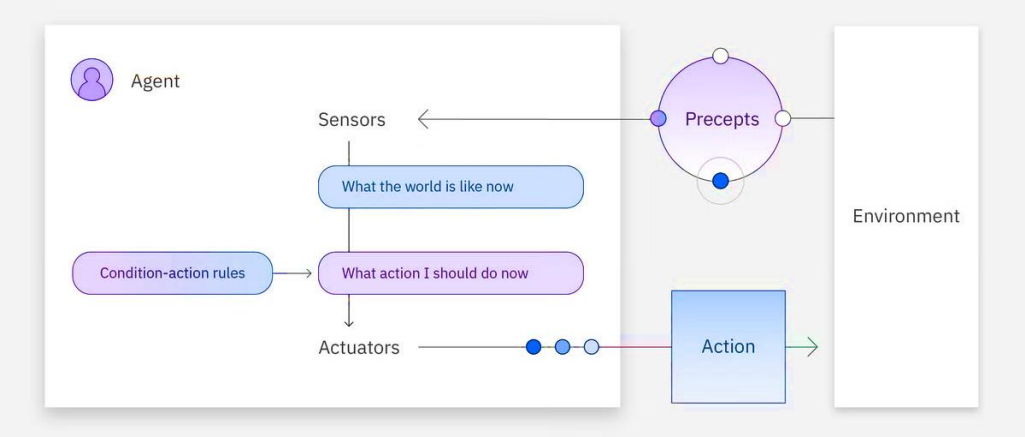


Types of AI Agents

Simple Reflex Agents

Simple reflex agents are the simplest agent form that grounds actions on perception. This agent does not hold any memory, nor does it interact with other agents if it is missing information. **[4]**



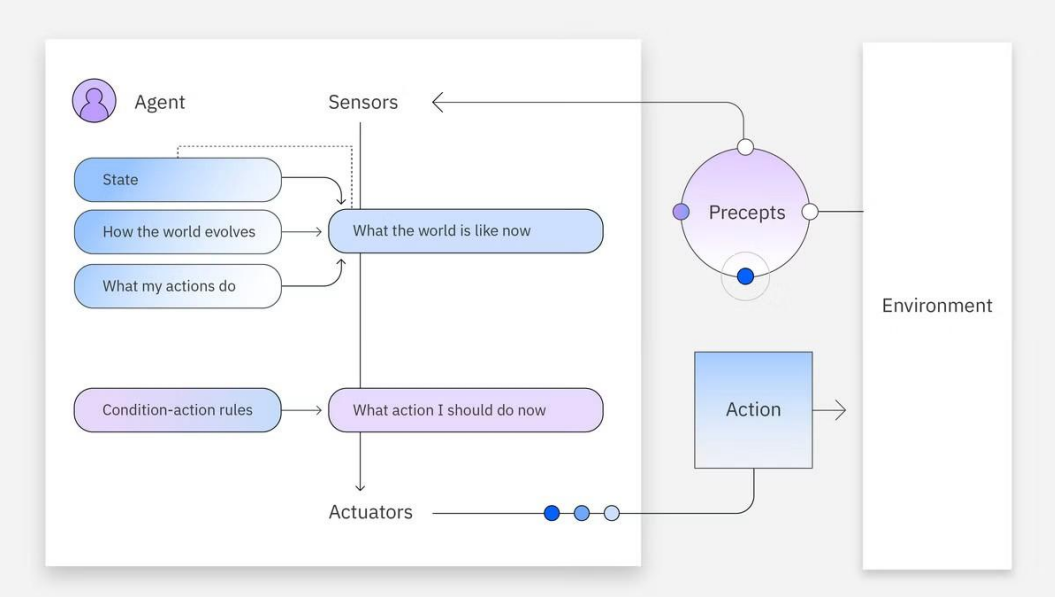
- No memory capacity
- Rule-based decisions
- Fast but inflexible

EXAMPLES

Auto doors, Thermostats, Motion lights

Model-Based Agents

Model-based reflex agents use both their current perception and memory to maintain an internal model of the world. As the agent continues to receive new information, the model is updated. The agent's actions depend on its model, reflexes, previous precepts and current state. **[4]**



- Maintains internal state
- Handles partial observability
- Updates knowledge continuously

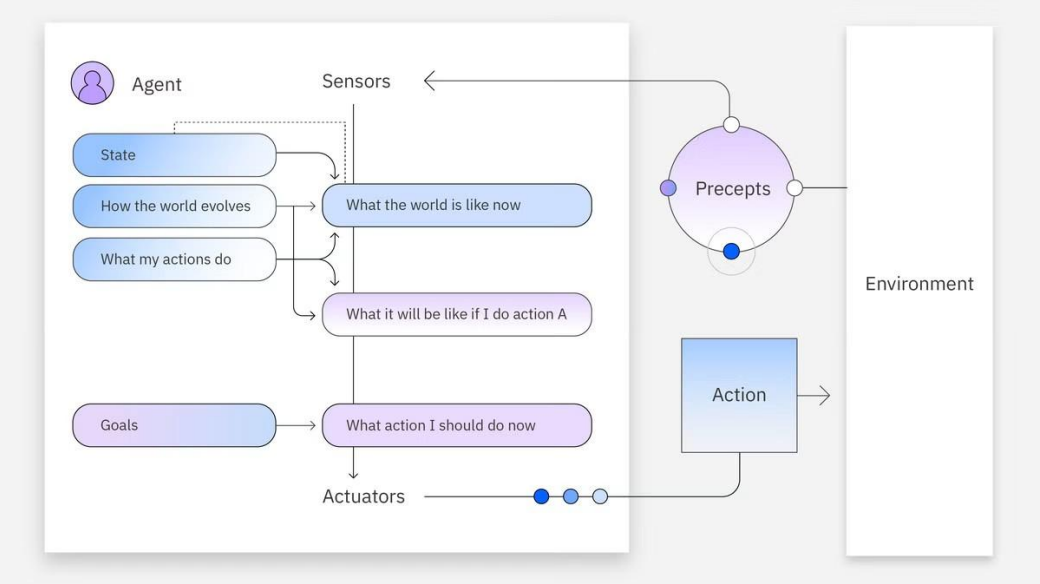
EXAMPLES

Robot vacuums, Warehouse robots

Types of AI Agents (Cont.)

Goal-Based Agents

Goal-based agents have an internal model of the world and also a goal or set of goals. These agents search for action sequences that reach their goal and plan these actions before acting on them. This search and planning improve their effectiveness when compared to simple and model-based reflex agents. **[4]**



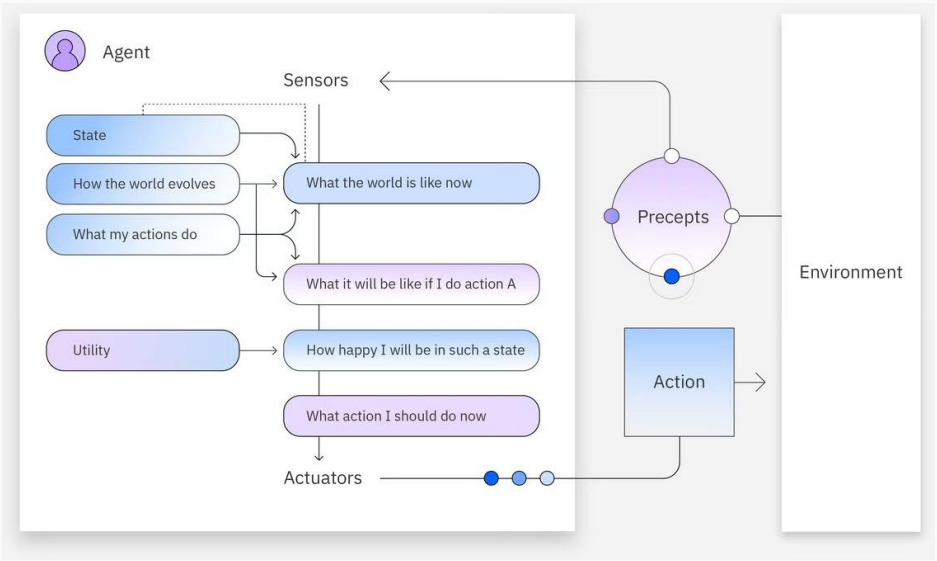
- Goal-oriented approach
- Planning & search capabilities
- Evaluates possible actions

EXAMPLES

GPS navigation, Logistics planners

Utility-Based Agents

Utility-Based Agents evaluate multiple possible actions using a utility function, selecting the option that maximizes overall performance and handles trade-offs efficiently. **[4]**



- Decision optimization
- Handles complex trade-offs
- Maximizes expected reward/utility

EXAMPLES

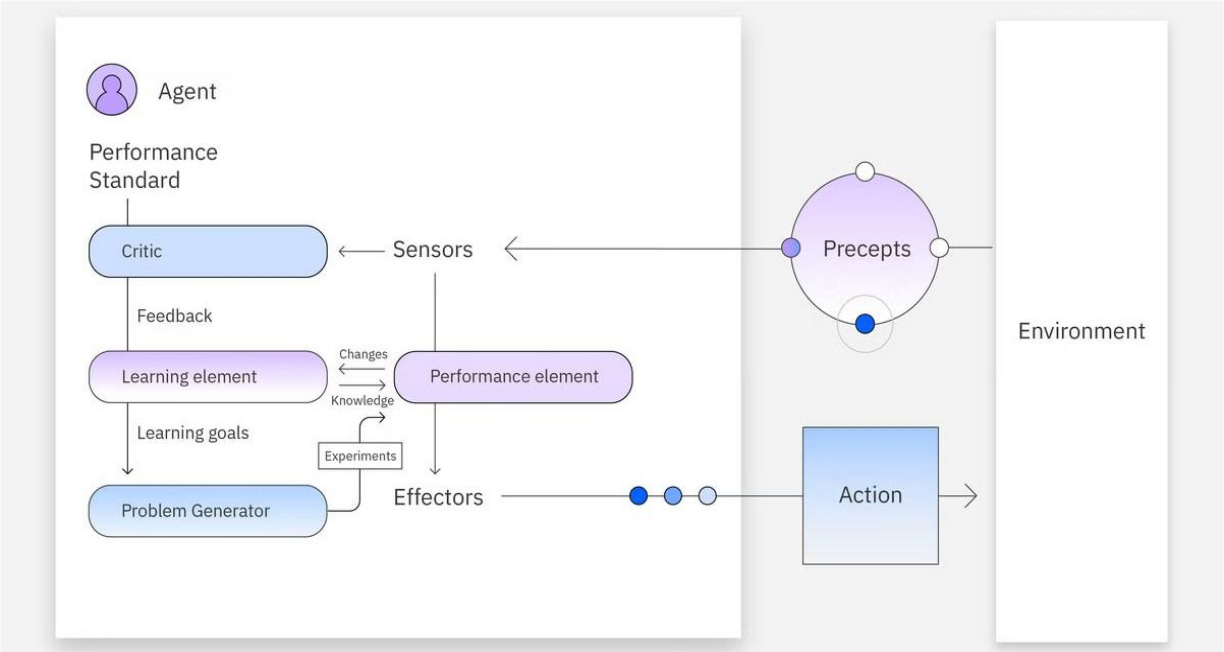
Navigation Systems

Types of AI Agents (Cont.)

Learning Agents

Learning agents hold the same capabilities as the other agent types but are unique in their ability to learn. New experiences are added to their initial knowledge base, which occurs autonomously. This learning enhances the agent's ability to operate in unfamiliar environments. Learning agents might be utility or goal-based in their reasoning and are composed of four main elements:

- **Learning:** This process improves the agent's knowledge by learning from the environment through its precepts and sensors.
- **Critic:** This component provides feedback to the agent on whether the quality of its responses meets the performance standard.
- **Performance:** This element is responsible for selecting actions upon learning.
- **Problem generator:** This module creates various proposals for actions to be taken. **[4]**



- Self-improving over time
- Learns directly from experience
- Adapts to changing environments

EXAMPLES

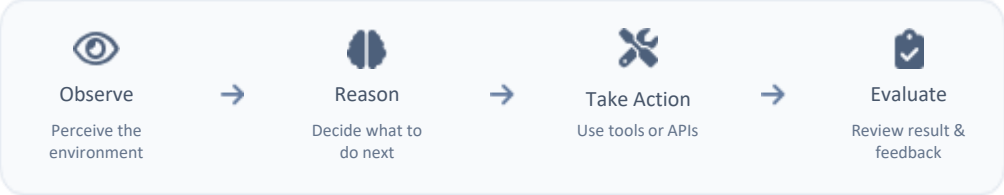
AlphaGo, Netflix Recommendations, Adaptive spam filters

AI Agent Architectures

Different patterns for designing agents based on how they reason, act, and collaborate.

1. ReAct (Reason & Act)

It combines reasoning and action in a dynamic, iterative loop where a single agent observes the environment, formulates a plan, executes an action using available tools or APIs, and evaluates the resulting feedback to decide the next step. By constantly re-evaluating its context, the model can course-correct on the fly, retry failed tool calls, or adjust its strategy based on intermediate outputs. [6]



Example Use Cases: Q&A, Code generation & debugging

2. Sequential Workflow

It arranges processing steps into a strict, deterministic assembly line where the structured output of one module directly feeds as the input to the next. Typical execution flows linearly through discrete stages: extracting raw data from a source, transforming and cleaning it into a structured schema, analysing key metrics, and generating a polished output. Because each phase has bounded boundaries, this architecture reduces latency and non-deterministic behaviour, making it ideal for standardized, step-by-step tasks

Example Use Cases: Document processing pipeline, Report generation

AI Agent Architectures (Cont.)

Different patterns for designing agents based on how they reason, act, and collaborate.

3. Routing Architecture

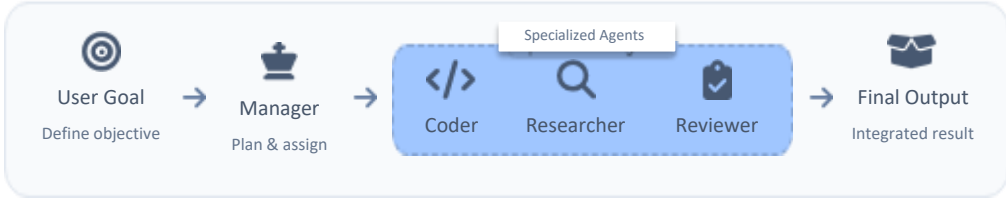
The **Routing Architecture** acts as an intelligent traffic controller that uses a centralized supervisor or classification node to interpret user intent before delegating execution. Once the router identifies the nature of the input query, it dynamically dispatches the request to a dedicated specialized sub-agent.

This separation keeps individual agent prompts lightweight, context windows focused, and token costs optimized, making it a natural fit for multi-domain customer support systems and triage-heavy enterprise applications.

Example Use Cases: Customer support automation

4. Multi-Agent Swarm

A **Multi-Agent Swarm** relies on a manager agent to orchestrate a ecosystem of domain-expert sub-agents working together toward a high-level goal. The manager breaks down complex user objectives into specific tasks and assigns them to specialized roles—such as a Coder for implementation, a Researcher for context gathering, and a Reviewer for quality assurance—who can debate, review each other's work, and iteratively refine deliverables. By enabling peer-level verification and specialized division of labour, swarms excel at tackling high-complexity, multi-layered domains



Example Use Cases: Software development teams, Threat analysis & support

What is MCP & Why Do We Need It?

- **The Core Problem:** LLM knowledge is frozen at training time & LLMs Agents cannot inherently interact with the outside world (real-time data, business systems, APIs).

- **The Solution (MCP):** The Model Context Protocol is an open standard that enables developers to build secure, two-way connections between their data sources and AI-powered tools. The architecture is straightforward: developers can either expose their data through MCP servers or build AI applications (MCP clients) that connect to these servers [18]

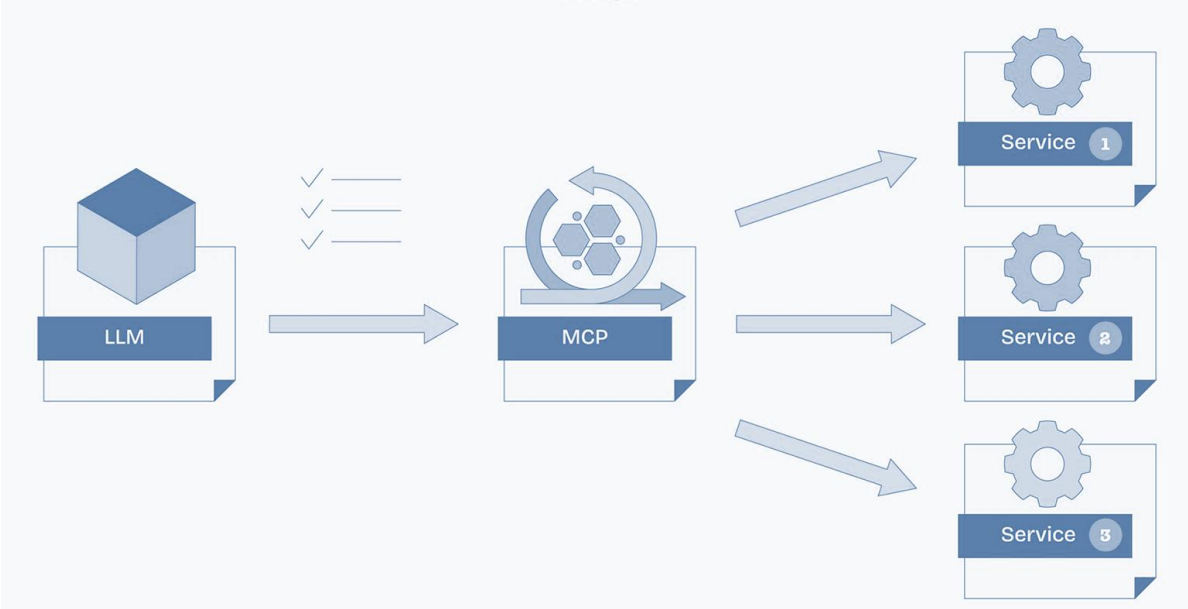
Here is a simplified look at how MCP would handle this:

1. Discover Tools: AI realizes it needs external help and finds the database and email tools registered on MCP.

2. Fetch Data: AI requests the sales report via MCP; the database server fetches and returns the report.

3. Send Email: AI hands the report to the email tool, which sends it to the manager.

4. Confirm: AI tells you the task is finished. [18]



The Traditional SDLC

The Traditional Software Development Life Cycle (SDLC) is the structured framework that engineering teams follow to plan, design, build, test, and deploy software applications from initial concept.

This Manual SDLC execution creates severe friction—engineers waste most of their time on administrative handoffs, environment setups, regression testing, and ticketing overhead instead of writing core logic.[5]

So, Our Aim is to automate the Traditional SDLC with Agentic AI, which will replaces these static bottlenecks with continuous, self-healing pipelines that autonomously execute tests, patch build failures, flag vulnerabilities, and manage releases. This **eliminates human error, compresses cycle times from months to hours, and frees development teams to focus on high-value architecture and innovation.**

Beyond Code Generation: Agentic AI

Strategic Comparison: Conversational AI vs. Agentic Systems. [16]

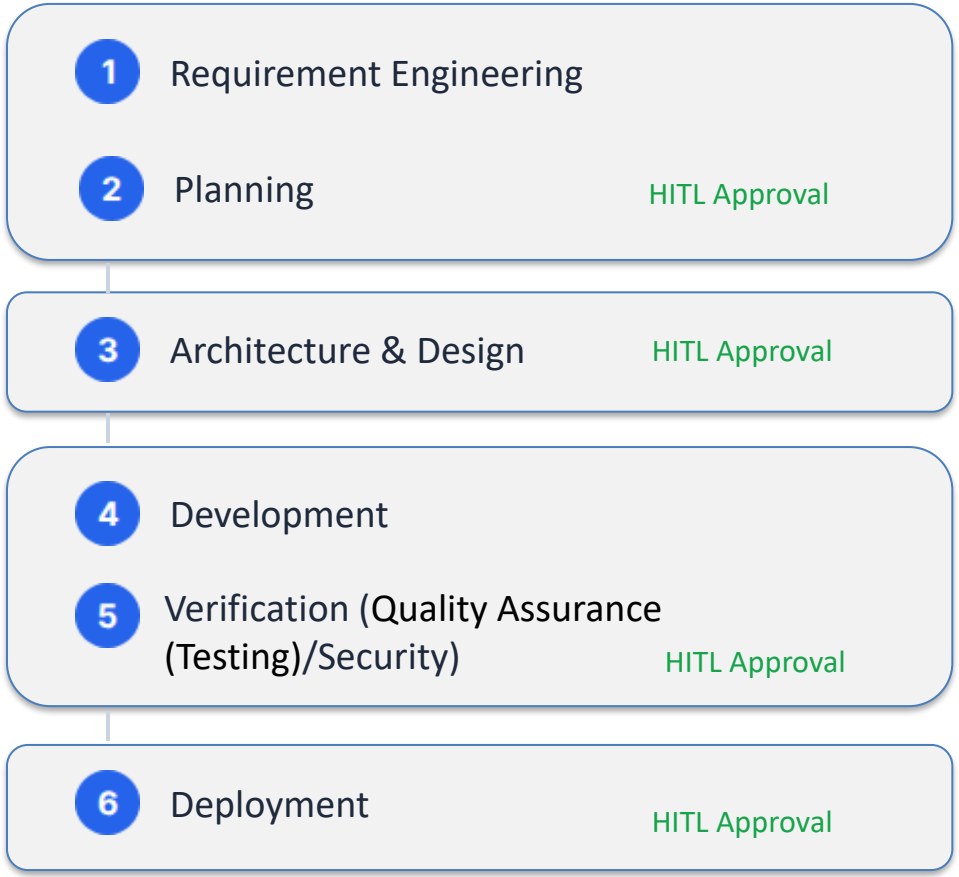
Feature / Aspect	AI Chatbot	AI Agent
Primary Function	Simulates conversation, often using rule-based or scripted responses to assist with simple Q&A.	Autonomous, goal-driven, and capable of reasoning to assist with multi-step tasks.
Personalization	Minimal personalization.	High personalization; adapts to user preferences over time.
Learning Ability	Limited learning ability.	Learns from user behavior and ongoing interactions.
Context Awareness	Low context awareness; struggles with complex tasks and context shifts.	High context awareness; handles complex, multi-step execution.
Operation Style	Provides passive, conversational assistance.	Provides proactive support to plan, create, and execute tasks.
Best Used For	Quick answers and basic customer service queries.	Complex problem-solving, everyday planning, and managing continuous tasks.

Agentic Frameworks

Framework	Execution Model	Core Strengths	Key Limitations	Ideal Use Cases	Offline / Local Capability
LangChain [7]	Chained Prompting & Tools	Rich toolset, extensive RAG ecosystem, vast API integrations	Workflow complexity at scale, abstractions can hide errors	General LLM applications, retrieval-augmented generation	**Excellent:** Natively connects to local LLMs (Ollama, Llama.cpp) and local vector stores (Chroma).
LangGraph [8]	Stateful Graph Orchestration	Cycles and loops, robust state persistence, time-travel debugging	Steeper learning curve, excessive setup for simple tasks	Production-grade multi-agent systems, human-in-the-loop control	**Excellent:** Inherits LangChain's local capabilities; operates entirely offline with self-hosted models.
CrewAI [9]	Role-based Team Collaboration	Easy setup, intuitive agent personas, automatic task delegation	Limited dynamic branching and complex workflow flow control	Autonomous research pipelines, content creation, structured teams	**Strong:** Explicitly supports local execution by binding agents to local endpoints via Ollama or LiteLLM.
OpenAI Agents SDK [10]	Lightweight Handoffs & Guardrails	Minimal abstractions, built-in tracing, standardized routing primitives	Lacks built-in persistent long-term memory or native graph structures	Rapid deployment of production agents, voice pipelines, clear delegation	**Very Good:** Provider-agnostic design allows it to run offline by routing to local models via the Chat Completions API.

Proposed Architecture: Modular Pipeline

A modular pipeline with integrated **Human-in-the-Loop (HITL)** governance gates to ensure enterprise safety.

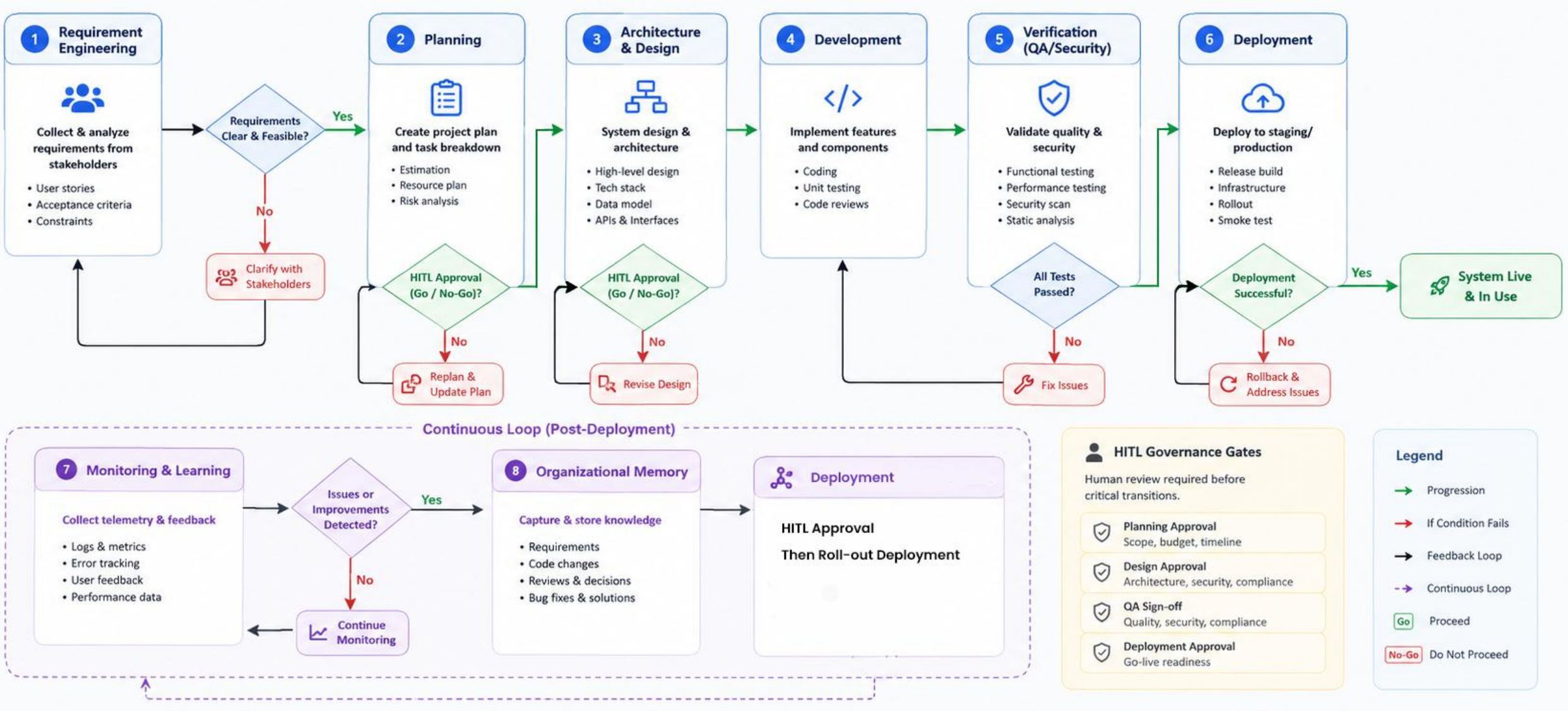


Continuous Loop

7. Monitoring & Learning: Consumes production logs, metrics, and user feedback to detect errors and optimize prompts.

8. Organizational Memory: Captures all requirements, code, reviews, and bug fixes into a Knowledge Graph for cross-sprint learning.

Proposed Architecture: Modular Pipeline



Identifying the Gap

While highly capable, existing autonomous systems index heavily on code generation, lacking end-to-end SDLC orchestration and organizational memory.

System	Requirements	Design	Coding	Testing	Deployment
Devin / OpenHands [14][15]	Partial	Partial	Yes	Yes	Partial
MetaGPT / ChatDev [11][12]	Yes	Yes	Yes	Partial	No
SWE-Agent [13]	No	No	Yes	Yes	No
Proposed System	Yes	Yes	Yes	Yes	Yes

Core Innovation: Adaptive Team Generation

Traditional: Fixed Agents

Standard agentic systems rely on rigid, hardcoded workflows (e.g., always Planner → Developer → Tester).

- Inefficient for small tasks (over-engineered).
- Lacks specialized roles (Security, DevOps) for complex enterprise apps. [1]

Proposed: Dynamic Generation

The Orchestrator analyzes project complexity, risk, and size before execution to spawn agents on-the-fly.

- **Small Script:** Spawns Planner + Developer.
- **Enterprise App:** Spawns Architect, Backend, Frontend, Security, QA, DevOps.
- Agents are retired when tasks complete, saving compute.

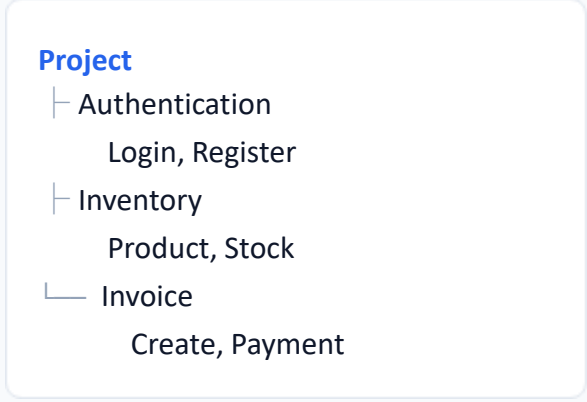
Core Innovation: Hierarchical Software Knowledge Graph (HSKG)

Beyond Traditional Memory:
A Context-Aware Project Memory

The Problem: Current AI agents retrieve entire repositories or large documents, resulting in high token consumption and irrelevant context. [1]

Our Solution: Our proposed HSKG stores the software project as a hierarchical graph, enabling agents to retrieve only the smallest relevant subgraph required for a task.

Project Graph



Purpose

- Represents the complete SDLC structure
- Maintains hierarchy between project components
- Enables modular navigation

Context-Aware

Instead of:
Entire Repository → LLM

Agent retrieves:

Login Module -> UI -> Backend Script & Variables -> API -> User DB

Benefits

- Minimal context
- Lower token usage
- Faster reasoning
- Better accuracy
- Reduced hallucination

Local Offline Execution Strategy

Deploying via offline models ensures data privacy. Model selection is based on VRAM footprint, Training data freshness, Suitable for Agentic Operations.

Model (4-bit Quant)	Recommended Hardware (VRAM)	Knowledge Cutoff	Agent Capability	Best Use Case
Gemma 4 4B	Laptop / iGPU (~5 GB)	Early 2026	⭐⭐⭐	Basic assistants, RAG, lightweight offline agents
Gemma 4 12B	RTX 4070 (8–10 GB)	Mid 2026	⭐⭐⭐⭐	General-purpose AI agents with good reasoning & tool use
Mistral Small 3.2 (24B)	RTX 4090 (16 GB)	Mid 2025	⭐⭐⭐⭐⭐	Enterprise agents, multi-tool workflows, planning
Qwen3-Coder 32B	Workstation / Mac M-Series (20–22 GB)	Mid 2025	⭐⭐⭐⭐⭐	Coding agents, debugging, software engineering
Llama 3.3 70B	Multi-GPU / High-End Server (48+ GB)	Late 2023*	⭐⭐⭐⭐⭐	Research, complex planning, multi-agent orchestration

Source: *Compiled from official model documentation and validated using publicly available implementation benchmarks and deployment experiences.*

Conclusion

We are proposing an **Autonomous Software Development Lifecycle (A-SDLC)** framework that transforms traditional software engineering into an **AI-Orchestrated Software Engineering Organization**.

Instead of isolated coding assistants, the system coordinates multiple specialized AI agents across the complete SDLC—from **requirements engineering** to **continuous monitoring and learning**.

Expected Outcomes

- Reduced context size and token consumption
- Faster and more accurate software development
- Improved traceability across the SDLC
- Persistent engineering knowledge

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[17] *IBM: https://www.ibm.com/think/topics/ai-agents*

[18] *Google Cloud*: *cloud.google.com/discover/what-is-model-context-protocol*

Thank You